



Environmental Activity-Based Costing and Its Role in Supporting Environmental Sustainability: An Applied Study at the State Company for Fertilizer Industry in Southern Basrah

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Abstract: This study aimed to demonstrate the application of environmental activity-based costing (EABC) and its impact on supporting environmental sustainability, in accordance with ISO 14001 and 14051 standards for material flow cost accounting (MFCA) and GRI 300 standards for materials, energy, water, compliance, waste, and environmental performance improvement. EABC is an environmental accounting tool that identifies activities and allocates environmental costs to those activities, then to products, thereby assigning each product its actual costs and providing more accurate data. The research was conducted at the General Company for Fertilizer Industries in the Southern Region of Basra, Iraq. The researcher employed a practical approach by comparing the system implemented in the company under study with EABC. The main reason for using this technique is the inefficient use of resources and the resulting environmental pollution and fines imposed for exceeding permissible pollution limits. These costs have come to constitute a large percentage of the company's total costs, thus impacting its profitability. The research contributed to identifying areas of waste resulting from the inefficient use of available resources and assisted management in making sound and accurate decisions related to environmental and economic aspects. It also helped improve environmental performance and enable the allocation of environmental costs to products based on their resource consumption. This, in turn, leads to the sustainability of resources through optimal use, thus achieving environmental sustainability. The study concluded that adopting cash flow statements helps improve various administrative decision-making processes, including pricing decisions, by allocating environmental costs to products and the activities that generate them. Furthermore, some reasons for waste in raw materials are attributed to the poor quality of those materials and the manual addition of materials. Therefore, the model directs management's attention and efforts towards purchasing less environmentally damaging materials and using a pump for material application.

Keywords: Environmental accounting; Activity-based costing; Environmental costs; Environmental sustainability; Environmental management accounting

1 Introduction

With the increasing challenges facing industrial companies worldwide due to technological development, product diversity, emissions, and the non-optimal use of resources, the need to focus on technology that addresses these negative aspects has emerged.

The scientific problem is that traditional methods suffer from many shortcomings in calculating costs more accurately and in addressing the environmental impact of industrial activities, resulting in the provision of inaccurate information, which in turn affects management decisions and the difficulty of management guidance towards sustainability.

The research problem can be identified with a central question: Does the implementation of the environmental activity-based (EABC) system contribute directly to the enhancement of managerial capacities?

Can adherence to ISO 14005, 14001, and GRI 300 standards reduce costs while enhancing the competitive advantage of industrial companies?

This study aims to achieve a balance between economic and environmental aspects to promote sustainability by understanding the reasons why industrial companies in Iraq fail to establish advanced accounting systems, despite the

urgent need for such systems in light of the global trend towards compliance with environmental protection laws.

The study seeks to clarify the possibility of allocating resources more accurately for each activity, thus accurately distributing environmental costs to different products to better reflect the consumption of resources and loading the product at its actual costs.

When applying the EABC technology, two effects are formed, the environmental impact works to reduce waste and waste and the optimal use of available resources, while the economic impact works to reduce costs, increase the profitability of the company and enhance its competitive advantage in a way that works to support environmental sustainability, and improve performance, with a focus on the General Company for the Fertilizer Industry sample for the study.

The main hypothesis is that the application of EABC technology contributes to the allocation of environmental costs and the determination of the environmental impact of activities in a way that ensures the efficiency of operational activities and enhances compliance with ISO 14005, 14001, and GRI 300 standards, which in turn supports environmental sustainability.

2 Literature Review

2.1 Conceptual Foundations

2.1.1 Activity-based costing

The Companies face a lot of pressure related to reducing their pollutants, but their accounting systems and strategic management struggle to solve this problem. Therefore, researchers have focused on finding accounting methods that provide accurate information that enables company management to identify the causes of pollution, waste locations, and pollutant costs, including that EABC is commonly used in modern energy management [1], where information about resource allocation and product costing is more clearly utilized. It is a powerful accounting method that helps track the costs of non-sustainable environmental products based on resource-consuming collaborations, leading to the production of integrated environmental products. Yang and Chang [2] considered activity-based costing (ABC) to be an accurate method for allocating small costs, as all resources, cost classification, and costing are combined to measure environmental costs. This helps assign costs to all products, enabling resource sustainability and innovations to gain a deep understanding of different disciplines, thus saving more costs [2, 3]. Hsieh et al. [4] also saw EABC as a two-stage accounting method that is an extension of modern cost accounting to increase the accuracy of calculating indirect costs and then tracing these costs through activities to products. Al Mashkoor [5] explained that EABC technology enables companies to neutralize the negative impact of pollutants and work to reduce them in order to improve environmental performance, and helps decision-makers develop strategies to enhance environmental performance.

2.1.2 Environmental costs

Environmental pollution is one of the increasingly dangerous phenomena as a result of industrial progress and the diversity of sources of pollution, as economic units seek to dispose of their waste in ways that often lead to serious damage to the environment, which reflects negatively on human health and the sustainability of natural resources [6]. The environment is the physical basis of human life, as it provides the necessary resources that ensure the survival and continuity of human development, giving it a vital and pivotal role in achieving the sustainability and progress of human life [7]. In traditional systems, environmental costs are incorporated into indirect industrial cost calculations. These costs are not systematically tracked and are not attributed to the relevant processes or products, which leads to them being hidden in overhead accounts [8]. Environmental accounting is one aspect of sustainability accounting that emphasizes the relationship between the economic unit and its impact on the environment [9]. Environmental accounting is understood as the management of financial, material, and qualitative information related to the environmental impacts and financial results of business activities related to the environment. This information supports internal and external decision-making, as well as reporting and accountability [10]. Environmental strategies significantly impact management accounting systems in organizations. When adopting positive interoperable strategies, such as the use of clean technology and improving production processes, they reduce environmental impacts. Environmental accounting systems are modified to provide accurate and comprehensive information about the environment [11]. Nguyen stated that environmental costs are collected in general expense accounts, meaning that environmental costs are hidden in the general expense account. Facts indicate that the accounting system in most companies hides many important environmental costs within general expense accounts [10]. Although environmental costs are an essential part of environmental accounting, previous studies have shown a lack of standards that accurately define these costs. This lack has generally appeared in the field of environmental management accounting, and accounting for intermediate costs in particular [12]. Environmental costs are defined as the costs incurred by the company as a result of its activities that affect environmental quality, with the aim of maintaining a clean and pollution-free environment, ensuring the protection of humans and animals, and maintaining the environmental balance in general [13]. The most prominent negative effects of not taking environmental costs into

account are the exacerbation of environmental problems to the point that production units may be forced to completely restructure and modify production technologies, weaken competitiveness, and deteriorate marketing performance. After measuring the environmental impacts resulting from the activities of the economic unit and integrating them into the accounting system [14], one of the main challenges in the field of accounting is the high percentage of environmental costs in some industries and the need to control them. Promoting environmentally friendly products enhances the competitive advantage in local and international markets [15, 16].

2.1.3 Environmental sustainability

Sustainability is an important topic that has received significant attention because it relates to the rational use of resources and energy while obtaining the greatest benefit from them, providing social and economic justice for all, and enhancing the ability to adapt to climate change. It is a method adopted by governments to achieve justice in the distribution of wealth between generations.

The concept of sustainability emerged in 1987 with the World Commission on Environment and Development in a study called the Brundtland Report, which relied on finding new ways of thinking about the meaning of sustainable development on a global level [17]. The principle of environmental sustainability is the basis for many strategies that drive companies to focus on improving environmental performance in addition to achieving financial profits. Environmental sustainability practices are not just policies or strategies, but rather a philosophy aimed at improving the environment by adopting environmentally responsible business practices, which support environmental sustainability and simultaneously provide greater profitability [18]. Environmental sustainability is defined as responsible interaction with the environment to avoid depleting or degrading natural resources and allowing for a long-term environmental balance. Adopting environmental sustainability helps ensure that the needs of the world's population are met today without compromising the ability of future generations to meet their needs [19]. Environmental sustainability has emerged through several key characteristics, the most important of which are the optimal use of resources and environmental conservation. Sustainability helps provide a healthy and clean environment and reduce pollution, which positively impacts the health and well-being of individuals [20].

To connect ABC with environmental sustainability, a technique must be used that connects these two variables. Therefore, the material flow cost accounting (MFCA) technique, according to ISO 14051, was used. This technique enables organizations to design an integrated environmental accounting information system that contributes to identifying opportunities for performance improvement, administrative decisions, and reducing overall costs through efficient resource management and the elimination of inefficient resource utilization. This enhances alignment between environmental and economic objectives and reinforces the concept of environmental sustainability for the organization [11]. This technique paves the way for the application of ABC to allocate each product to its actual environmental costs. After identifying areas of waste in materials, energy, water, and waste according to the GRI 300 standard and calculating environmental costs, the units responsible for these costs can reduce them or avoid practices that lead to this waste and these costs.

2.2 Previous Empirical Applications

Özçelik [20] analyzed the integration of relationship-based costing and micro-cost accounting to optimize costs and minimize impacts, aiming to create a better management structure and reduce expenses by applying this integration within the corporate community. The study concluded that the most important aspect is identifying the smallest costs, which are not considered investments but rather part of general expenses. Companies should identify these costs more frequently and in greater diversity to help them manage them more effectively. This enables them to manage small and micro-sized expenses, allowing them to focus on environmental improvement, social responsibility, and reporting to auditors. This technique empowers them to develop their businesses and improve their market and organizational standing.

Therefore, companies must be proactive, integrate their environmental policies into national policies, and report on their environmental performance to auditors. All of this contributes to the sustainable development of the central and marketing corporate structure.

Study by Tsai et al. [21] aimed to balance environmental and economic objectives in the aluminum wheel industry by using green ABC models and applying them to carbon emission management. The study focuses on optimizing environmental costs while maintaining companies' financial performance amidst changing environmental policies, such as carbon taxes and trading mechanisms. The research found that integrating carbon trading mechanisms with tax reductions can improve companies' financial position without significantly impacting environmental objectives. It is possible to achieve a balance between reducing carbon emissions and maximizing profits by implementing flexible regulatory strategies for carbon taxes. Identifying the optimal points for balancing environmental and economic objectives is a complex process that requires the use of advanced mathematical models to achieve sustainability. The study recommends developing flexible and complex tax policies that adapt to market fluctuations and help companies achieve a balance between environmental and economic objectives. Companies should adopt new technologies such as Industry 4.0 and apply circular economic principles to improve production sustainability and reduce their carbon

footprint. Companies should be encouraged to integrate carbon emission management strategies with production strategies to reduce environmental costs and maximize profits. The study recommends investing in research and development to develop low-carbon production technologies and achieve global emission reduction aims.

Study by Rahman et al. [22] aimed to examine the impact of environmental management systems and environmental accounting on achieving sustainability. It indicates that integrating effective environmental practices with operational costs can lead to a sustainable competitive advantage. By reducing negative environmental impacts and lowering environmental costs, these practices enhance the economic and environmental sustainability of companies. The study concludes that environmental sustainability can be a major source of competitive advantage rather than a financial burden. This makes it essential for companies in developing economies to adopt environmental accounting as part of their strategies to ensure long-term financial and environmental sustainability. The study recommends supporting the integration of environmental accounting into corporate strategies to achieve sustainable efficiency, which will help reduce costs and support sustainability. It also encourages companies to invest more in environmental management technologies to achieve environmental and financial benefits. Furthermore, there is a need for government strategies that require companies to adhere to sustainable environmental practices to achieve a balance between economic development and environmental protection.

2.3 Research Gap and Rationale for the Present Study

2.3.1 Similarities and differences between previous studies and the current study

(1) Key contributions of previous studies

Previous studies have strengthened the current study by assisting the researcher in taking a lateral view, providing guidance, information, suggestions, and value regarding cost accounting and cost-effective management. The EABC technique would not have been used at the level of economic units were it not for such studies. Furthermore, previous studies have significantly contributed to identifying key scientific and practical points for the researcher in integrating mathematical management accounting with smart electronics to improve the identification of lower costs. They have also contributed to the flow of resources, encouraging learning how to utilize resources and reducing the minutiae of multiple costs. This facilitates decision-making that contributes to achieving economic, social, and environmental sustainability, as well as helping improve other aspects. What matters most is the need to move from theoretical to practical application of small-scale accounting, and for accountants to have the knowledge to carefully identify small contributions and the costs they incur.

(2) Research gaps

Previous subscriptions were not determined until the direct cost system was not defined from small costs and accurately measured, as accurate data determines the original system, which requires significant effort and time, or complementary. These beautiful costs were not determined separately, but the hidden costs were not calculated. Previous Arab and foreign studies remain in their interpretation of the multiple activities and integrated costs within integrated management systems. Some of you are determining the direct effects and integrating them within the product cost, only for ABC, to learn electronically in a system of precise determination. This clear determination of the complete origin of these concepts determines the undefined information that deals with the process of making the joint work, which does not confirm further research and clarification, because it is accurate and comprehensive for the environmental and economic dimensions separately.

(3) Contributions of this study

The current study focuses on EABC, a technique developed from the traditional activity-based costing system due to the growing emphasis on environmental issues. This study incorporates the theoretical and scientific foundations of EABC, including its concept, rationale, objectives, and steps. It demonstrates how costs are calculated based on the environmental activity of each product, to transform the company's products into environmentally friendly ones. The study also examines the impact of EABC on supporting sustainability by converting products from traditional to environmentally sustainable ones that adhere to environmental standards. The study was conducted in a real industrial setting, specifically the General Company for Southern Fertilizer Industries in Iraq, thus supporting its practical application. Furthermore, the study provides a precise allocation of environmental costs across production units and links them to the final product (urea fertilizer), thereby improving cost accuracy and enhancing decision-making.

2.3.2 Statement of the validity of previous studies

Previous studies strengthened the current study by assisting the researcher in the theoretical aspect, providing guidance, information, suggestions, and ideas. Documentary costing techniques related to environmental activities were not used at the level of economic units without following up on these studies.

Previous studies also significantly contributed to students' understanding of the scientific and tourism perspectives regarding the integration of electronic management accounting with traditional electronic accounting to improve the identification of micro-costs.

Providing micro-cost data for children in general supports decision-making that contributes to achieving economic, social, and environmental sustainability. It also helps contribute to contributing to contributing to contributing to contributing to achieving economic, social, and environmental sustainability.

The study highlighted the need to move from the theoretical to the practical application of micro-accounting and the importance of accountants knowing how to accurately determine the benefits and expenses of micro-accounting.

3 Methodology

3.1 Research Method

The researcher followed descriptive models to separate the distance and the best to achieve the research objectives, by combining theory through a review of previous studies with the applied side case in the General Company for Rapid Southern Industry.

3.2 Data Collection

The study relied on available literature, including books, theses, and dissertations, as well as research and scientific articles published in journals and websites, in both Arabic and English. Unstructured interviews were conducted with specialists at various levels and departments related to the research topic. The observation method was used through field visits to follow the progress of the production process and to view work sites. Records, documents, and reports related to the research sample were reviewed.

3.3 Data Analysis

Environmental costs were analyzed using the following detailed process. First, cost data was obtained from a conventional costing system. Second, MFCA was applied to track material and energy flows and identify waste. Third, EABC was used to allocate environmental costs to economic activities and then to final production outputs.

The environmental cost was calculated using the following general formula:

$$\text{Environmental cost} = \text{Waste} \times \text{Cost per unit}$$

Based on this approach, environmental costs were identified, measured, and allocated to production units, and ultimately linked to the final product (urea).

4 Results

4.1 Determining the Quantities of Waste and Calculating Environmental Costs

Material flow technology is used according to ISO 14051, which monitors the flow of materials within each stage of the production process. Actual inputs, to determine the amount of waste or savings, to determine the environmental impact, to improve performance, and to improve company decisions.

4.1.1 Water unit

Initially, the water is treated in this unit after it is pumped from the Mahila station, then the water passes through the precipitation and filters to remove any impurities stuck in the water, as an initial filter, and then aluminum sulfate as a coagulant. Poly-electro-willed to enlarge the suspended particles and enhance the sedimentation process, and some chemicals to turn into filter water, part of which, the largest part, goes to the cooling unit to cool the heat exchangers of the steam boiler unit and the ammonia unit, and the urea unit. The other part of the water is treated after adding sodium sulfite to it to remove chlorine with water and EDTA to wash the reverse osmosis membranes, sodium phosphate to prevent calcification, and sulfuric acid to prevent the deposition of salts. And activate the membrane of reverse osmosis water and some other materials, to turn it into water free of ions, to go to the steam boiler unit to produce steam, which is used in the ammonia and urea units. The Quantity of waste in the water unit was in some substances, including aluminum sulfate (346.534925) t, poly-electrolyte (0.735365) t, sodium sulfate (0.376308), and phosphate. Sodium (2.92352), sodium hydroxide (30.5), chlorine gas (10.8) t, sulfuric acid (31) t, EDTA (3) t. Aluminum sulfate is 88% of the total amount consumed, and this is a very high percentage, which must be reduced according to the optimal use of resources in a way that ensures the achievement of environmental sustainability. And the environmental costs amount to IQD 410.92 million.

The environmental costs for each material were calculated using the following equation: (Quantity of waste × cost per unit).

- $346.534925 \times \text{IQD } 792.00 \text{ thousand} = \text{IQD } 274.46 \text{ million}$
- $0.735365 \times \text{IQD } 13.37 \text{ million} = \text{IQD } 9.83 \text{ million}$
- $0.376308 \times \text{IQD } 5.94 \text{ million} = \text{IQD } 2.24 \text{ million}$
- $2.92352 \times \text{IQD } 5.45 \text{ million} = \text{IQD } 15.92 \text{ million}$
- $30.5 \times \text{IQD } 1.29 \text{ million} = \text{IQD } 39.45 \text{ million}$

- $10.8 \times \text{IQD } 1.10 \text{ million} = \text{IQD } 11.86 \text{ million}$
- $31 \times \text{IQD } 528.00 \text{ thousand} = \text{IQD } 16.40 \text{ million}$
- $3 \times \text{IQD } 13.37 \text{ million} = \text{IQD } 40.76 \text{ million}$

Then, the environmental costs for each material within a unit of water were summed to obtain the total cost per unit of water.

• $274.46 \text{ million} + 2.24 \text{ million} + 9.83 \text{ million} + 15.92 \text{ million} + 39.45 \text{ million} + 11.86 \text{ million} + 16.40 \text{ million} + 40.76 \text{ million} = \text{IQD } 410.92 \text{ million}$

4.1.2 Cooling tower unit

In this unit, a portion of the filtered water is used to cool the heat exchangers of the units. This is done through a network of pipes extending underground, directed to the units that use the chilled water, and the Quantity of waste in this unit was equal to 10.316976 t in chlorine gas, and the environmental costs amounted to IQD 11.30 million.

- $10.316976 \times \text{IQD } 1.10 \text{ million} = \text{IQD } 11.30 \text{ million}$

4.1.3 Steam boiler unit

In this unit, deionized water is used. It is heated to a high temperature inside the steam boiler, converting into steam that circulates in a closed loop. Part of the steam goes to the ammonia unit, and the remainder to the urea unit. The Quantity of waste in this unit is 3,335.14 t for hydrazine and 2,280.559 t for hexavalent sodium phosphate. The environmental costs amounted to IQD 6.81 billion.

- $3,335.14 \times \text{IQD } 845.00 \text{ thousand} = \text{IQD } 2.82 \text{ billion}$
- $2,280.559 \times \text{IQD } 1.75 \text{ million} = \text{IQD } 3.99 \text{ billion}$
- $\text{IQD } 2.82 \text{ billion} + \text{IQD } 3.99 \text{ billion} = \text{IQD } 6.81 \text{ billion}$

4.1.4 Ammonia unit

In this unit, liquid ammonia is produced, which is used in the subsequent unit, the urea unit. During the use of a group of chemicals at specific temperatures and pressures to produce liquid ammonia, there are significant wastes in some of the materials used, including reaction gas (38678664) m³, combustion gas (34166662) m³, steam (725456) t, and electricity (6569400) kW. Environmental costs amounted to IQD 4.30 billion.

- $38678664 \times \text{IQD } 2.14 = \text{IQD } 82.84 \text{ million}$
- $34166662 \times \text{IQD } 2.14 = \text{IQD } 73.17 \text{ million}$
- $725456 \times \text{IQD } 33.48 = \text{IQD } 219.93 \text{ million}$
- $6569400 \times \text{IQD } 5.41 \text{ thousand} = \text{IQD } 3.93 \text{ billion}$
- $82.84 \text{ million} + 73.17 \text{ million} + 219.93 \text{ million} + 3.93 \text{ billion} = \text{IQD } 4.30 \text{ billion}$

4.1.5 Urea unit

In this last unit, ammonia produced from the ammonia unit and steam produced from steam boilers are used. Some gases produced as byproducts in the ammonia unit are added, along with other materials. Under specific pressure and temperature, urea granules are produced as a fertilizer and sold to the Agricultural Supplies Company. The wastes in this unit amount to 15,275.62 t in ammonia, 224,980.7 t in carbon dioxide, 980,528 kW in electricity, and 383,100.2 t. In water vapor, the environmental costs in this factory amounted to IQD 2.35 billion.

- $15,275.62 \times \text{IQD } 15.99 \text{ thousand} = \text{IQD } 244.27 \text{ million}$
- $224,980.7 \times 0 = 0$
- $980,528 \times \text{IQD } 33.48 = \text{IQD } 32.83 \text{ million}$
- $383,100.2 \times \text{IQD } 5.41 \text{ thousand} = \text{IQD } 2.07 \text{ billion}$
- $244.27 \text{ million} + 32.83 \text{ million} + 2.07 \text{ billion} = \text{IQD } 2.35 \text{ billion}$

4.2 Calculating the Company's Total Environmental Costs

Therefore, the total environmental costs in the fertilizer company for the production units amounted to IQD 13.88 billion. As for the other units that bear part of the environmental costs according to a rate of 13%, it amounts to IQD 2.58 billion. Therefore, the total environmental costs in the company amounted to IQD 16.46 billion, which represents 15.6% of the company's total costs. When applying ABC, environmental costs are allocated by identifying activities, assigning allocation rates to each activity, determining the necessary guidelines, and then distributing these costs to the activities that generate them. These costs are then allocated to the products. This process identifies areas of waste, allowing for their avoidance, and pinpoints the primary cause of these costs to eliminate or reduce them. Consequently, the company realizes a profit equivalent to the environmental costs. For example, the ammonia production cost is IQD 45.79 billion after deducting environmental costs of IQD 4.30 billion. Similarly, the urea plant's economic costs are IQD 42.99 billion after deducting environmental costs of IQD 2.35 billion. Therefore, the cost per ton of the final product, urea fertilizer, becomes IDQ 371,668, down from IDQ 440,000, representing a profit for the company.

Through the production process, which goes through several stages, some chemical materials are added manually due to a malfunction in the pump that pumps the materials in the previous units. This leads to the addition of more

than specified. Therefore, the standard inputs for production are compared with the actual quantities to determine the difference, whether it is a savings in materials, an equal amount, or a waste of materials. Consequently, the quantities of waste will be calculated for environmental costs by multiplying them by the price per unit. Therefore, the company should adopt EABC as it is a good solution for both economic and environmental aspects. It works to reduce the negative environmental impact by encouraging companies to optimize resource use, meaning the efficiency with which they consume resources according to actual needs. This, in turn, helps preserve natural resources without wasting materials, energy, and water, thus ensuring future generations have access to these resources and achieving environmental sustainability. Furthermore, it reduces emissions of toxic gases such as CO₂, which harm the environment. Adherence to environmental laws is essential to minimize negative environmental impact and improve performance. From an environmental perspective, this will reduce costs in the form of environmental costs, as this technology identifies areas of waste and losses in each unit. Consequently, each unit can reduce the amount of wasted resources. Additionally, fines and taxes resulting from non-compliance with environmental regulations and laws, such as ISO 14001 and ISO 14051, will decrease. ISO 14051 emphasizes tracking material flows to minimize waste through the use of MFCA in research. These results also align with the retirements of the GRI (300-Series) standard. This strengthens the link between the results of applying this technology and the theoretical aspects of environmental management accounting and environmental sustainability. This, in turn, will give the company a competitive advantage and ensure its long-term market viability.

4.3 The Amounts of Waste and Environmental Costs of the Final Product in the General Company for Industry Fertilizers/Southern Region

4.3.1 Waste quantities for the urea plant for the year 2021

Table 1 shows the quantities of waste generated in the urea plant for the year 2021.

Table 1. Waste quantities for the urea plant

Months	Water Steam (t)		Electricity (kW)		CO ₂ (t)		Ammonia (t)	
	Urea 2	Urea 1	Urea 2	Urea 1	Urea 2	Urea 1	Urea 2	Urea 1
January	34224.58	0	0	0	23240.98	0	1741.408	0
February	28652.3	0	118560	0	23114.13	0	1364.28	0
March	14878.3	0	257160	0	15988.13	0	977.08	0
April	30665.32	0	173604	0	25838.9	0	1274.832	0
May	36991.96	0	0	0	17048.2	0	1510.696	0
June	29742.06	0	3832	0	15713.58	0	1190.256	0
July	24121.96	0	204912	0	12424.2	0	1074.496	0
August	30364.3	0	222460	0	18848.88	0	828.68	0
September	45496.6	0	0	0	21824.5	0	1138.16	0
October	34511.06	0	0	0	19998.58	0	1266.656	0
November	34818.2	0	0	0	14216.25	0	1470.52	0
December	38633.52	0	0	0	16724.4	0	1438.552	0
Total	383100.2	0	980528	0	224980.7	0	15275.62	0

Note: Cost data was obtained from a traditional costing system and then analyzed by the researcher using material flow cost accounting (MFCA) to optimize resource planning and identify material waste in each production unit. This was achieved by calculating the difference between actual and standard inputs, thus determining that any amount exceeding the standard quantities wasted. Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

By looking at the material flow statements for the months As shown in the Table 1, we find that the company under study hurts the environment and climate through the large amounts of waste in each of the raw materials, energy and water, as each of them has a different impact on the environment or even on the economic system, whether when extracting or obtaining the raw materials, or whether it is through waste and waste, or even in the form of energy, where it is possible to clearly imagine the resulting waste of energy and the accompanying waste of gas and air, as it is not offset by any benefit. This material model shows the reality of electricity consumption, water vapor, and CO₂ gas, and what are the potential future environmental impacts and the amount of negative damage to the environment. By simply looking at the amounts of waste, we can know the size of the loss in material productivity, as it shows that the outputs are less than the inputs because the inputs = actual products + the quantity of waste.

4.3.2 Calculation of waste costs (environmental costs) for the urea plant for the year 2021

As shown below, Table 2 presents the calculation of environmental costs for the urea plant for the year 2021.

Table 2. Calculation of environmental costs in the urea plant for the year 2021

Details	Waste Quantities			
	Water Steam	Electricity	CO ₂	Ammonia
Total waste quantities	383100.2 (t)	980528 (kW)	224980.7 (t)	15275.62 (t)
Price of unit	IQD 5.41 thousand	IQD 33.48	0	IQD 15.99 thousand
Waste costs (environmental costs) per item	IQD 2.07 billion	IQD 32.83 million	0	IQD 244.27 million
Total		IQD 2.35 billion		

Note: Environmental costs were then calculated by multiplying the waste by the price set by the company. Environmental costs = Price of unit × Total waste quantities. Environmental costs ammonia = $15.99 \times 15275.62 = \text{IQD } 244.27$ million. Waste cost ratio = Waste cost per item / Total waste costs. Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

It appears that the percentage of waste in costs is 82%, and the failure to optimally utilize this large quantity in production negatively impacts the company in terms of increased costs and reduced profitability, in addition to the negative impact in terms of the scarcity of some resources. Meanwhile, the percentage of waste in environmental costs for ammonia and electricity is negligible.

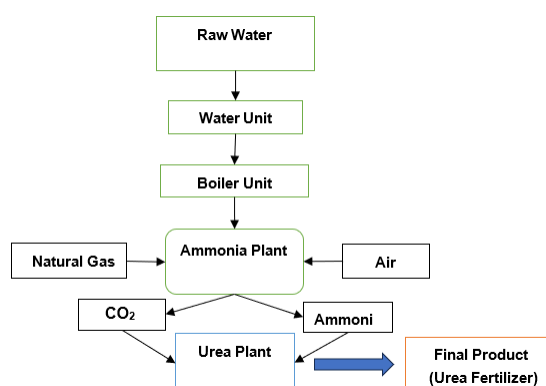
The total environmental costs for the five units equal IQD 13.88 billion, and IQD 2.58 billion for the production services units amounted to 13% of the total cost of these units to make the total equal IQD 16.46 billion.

4.4 Applying Document Technology to Environmental Activity within the Company

After identifying environmental costs, these costs should be allocated using the EABC technique. and the company relies on a production phase system to complete its final product, urea fertilizer. These phases are as follows:

- Step1: Water treatment
- Step2: Cooling unit
- Step3: steam production
- Step4: Liquid ammonia and carbon dioxide production
- Step5: The final phase, urea fertilizer production

As shown in Figure 1, the production process flow of the Southern Region Fertilizer Company is presented.

**Figure 1.** The production process flow in the Southern Region Fertilizer Company

4.4.1 Determining the driver and the driver amount

Table 3 shows the determination of the cost driver and its amount.

Table 3. Determination of cost driver and its amount

Activities	Type of Activities	Driver	Driver Quantity	Urea	Ammonia
Supporting activities	Water treatment unit	Quantity of received water (m ³)	17332030	6803827.953	10528202.05
	Cooling towers	Quantity of cooling water consumed (m ³)	7841512	3078248.673	4763263.327
	Steam production	Quantity of steam produced (t)	1839906	722270	1117636
Primary activities	Ammonia production	Quantity of ammonia produced (t)	163406	156676	6730
	Urea production	Quantity of urea produced (t)	238852	238852	0

Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

4.4.2 Calculation of the loading rate

Table 4 shows the calculation of the loading rates.

Table 4. Calculation of the loading rates

Unit/Plant	Quantity Driver	Environmental Costs	*Rate
Water unit	17332030 (m ³)	IQD 410.92 million	IQD 23.71
Cooling unit	7841512 (m ³)	IQD 11.30 million	IQD 1.44
Boiler unit	1839906 (t)	IQD 6.81 billion	IQD 3.70 thousand
Ammonia plant	163406 (t)	IQD 4.30 billion	IQD 26.33 thousand
Urea plant	238852 (t)	IQD 2.35 billion	IQD 9.84 thousand

Note: *Loading rate = Environmental costs / Quantity driver. Water treatment unit = IQD 410.92 million / 17332030 m³ = 23.70 (IQD). Similarly, for the remaining activities. Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

4.4.3 Allocation of indirect environmental costs to products

Table 5 shows the allocation of environmental costs to the product.

Table 5. Allocation of environmental costs to product

Types Costs	Categories Activity	Quantity Driver		Rate	Cost Production	
		Urea	Ammonia		Urea	Ammonia
Supporting activities	Water treatment unit	6803827.953 (m ³)	10528202.05	IQD 23.71	IQD 161.31 million	IQD 249.61 million
	Cooling unit	3078248.673 (m ³)	4763263.327	IQD 1.44	IQD 4.44 million	IQD 6.87 million
	Boiler unit	722270 (t)	1117636	IQD 3.70 thousand	IQD 2.67 billion	IQD 4.14 billion
Primary activities	Ammonia plant	156676 (t)	6730	IQD 26.33 thousand	IQD 4.12 billion	IQD 177.22 million
	Urea plant	238852 (t)	0	IQD 9.84 thousand	IQD 2.35 billion	0

Note: Environmental cost per product = Loading rate × Activity driver. Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

4.4.4 Total environmental costs per product

As shown below, Table 6 presents the total environmental costs per product.

Table 6. Total environmental costs per product

Details	Urea	Ammonia	Total
Hidden environmental costs within the units	IQD 9.31 billion	IQD 4.57 billion	IQD 13.88 billion

Note: Total environmental costs are the sum of the environmental costs for each of the five units above. Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

4.4.5 Total direct cost and indirect costs, economic and environmental

Table 7 shows the total company costs, including environmental costs, as presented below.

As shown in the Table 7, the company's total costs for ammonia and urea producers, including environmental costs, which equal IQD 105.23 billion. When environmental costs are excluded from the total costs, the producers' costs decrease by the amount of the environmental costs, as shown in Table 8.

Table 7. Total company costs, including environmental costs

Details	Urea	Ammonia	Total
Direct costs	IQD 1.64 billion	IQD 8.42 billion	IQD 10.06 billion
Indirect economic costs	IQD 41.35 billion	IQD 37.37 billion	IQD 78.71 billion
Indirect environmental costs of activities	IQD 9.31 billion	IQD 4.57 billion	IQD 13.88 billion
Total costs	IQD 52.30 billion	IQD 50.36 billion	IQD 102.65 billion
Hidden environmental costs within non-productive units			IQD 2.58 billion
Total			IQD 105.23 billion
Production quantity (t)		163,406	238852
Total cost per unit		IQD 308.18 thousand	IQD 440.57 thousand

Note: Economic indirect costs = Total indirect costs – Environmental indirect costs + Marketing and administrative costs. Indirect economic costs urea = 37.00 – 9.31 – 2.58 + 16.24 = 41.35 billion IQD. Similarly for Ammonia. Total costs = Direct cost + Indirect economic costs + Indirect environmental costs of activities + Hidden environmental costs = 10.06 + 78.71 + 13.88 + 2.58 = IQD 105.23 billion. Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

Table 8. Total company costs after excluding environmental costs

Details	Urea	Ammonia	Total
Direct costs	IQD 1.64 billion	IQD 8.42 billion	IQD 10.06 billion
Indirect economic costs	IQD 41.35 billion	IQD 37.37 billion	IQD 78.71 billion
Total costs	IQD 42.99 billion	IQD 45.79 billion	IQD 88.77 billion
Production quantity (ton)		163406	238852
Total cost per unit		IQD 280.21 thousand	IQD 371.67 thousand

Source: Prepared by the researcher based on the study data.

Therefore, this comparison shows that identifying waste and losses and determining their environmental costs enables the company to improve its performance and reduce costs in the ammonia plant from IQD 308.18 thousand per ton to IQD 280.21 thousand, and in the urea fertilizer plant from IQD 440.57 thousand per ton to IQD 371.67 thousand, and increase its profitability by the amount of environmental costs (IQD 13.88 billion) after eliminating them, strengthening its position in the market, and also leading to a decrease or absence of fines as a result of compliance with environmental laws. This is from the economic aspect. As for the environmental aspect, it encourages the company to make optimal use of resources according to a specified amount, which will reduce harmful emissions to the environment, which in turn works to support environmental sustainability in preserving natural resources and the share of future generations of those resources and avoiding their scarcity.

4.4.6 Partial income statement

Table 9 shows the company's partial income statement, illustrating the impact of the environmental cost allocation process on product profitability.

Table 9. The company's partial income statement to demonstrate the impact of the environmental cost allocation process on the level of product profitability

Details	Urea	Ammonia
Sales price	IQD 440.57 thousand	IQD 440.57 thousand
Total cost per unit	IQD 371.67 thousand	IQD 280.21 thousand
Profit	IQD 68.33 thousand	IQD 159.79 thousand
Profit ratio	17%	36.3%

Note: Profit ratio = Profit / Sales price. Source: Prepared by the researcher based on data from the General Company for Southern Fertilizer Industry.

From the Table 9, it can be seen that producers are profitable. By comparing the results extracted from the lists prepared according to the EABC system with the results extracted according to the traditional system, and determining the amount of reduction, we can observe the following:

Note that there is a difference between the two systems. In the traditional system, the cost of a single produced unit is high, as in the traditional system, environmental costs are hidden within indirect manufacturing costs. This has led to inaccurate costing of products, thus affecting management decisions. However, when applying the EABC system, we note that the cost of the product is reasonable, as the environmental costs have been accurately identified and the areas of waste and loss are known. This, in turn, makes it possible to reduce costs for products. When

applying the activity-based costing technique, the environmental costs are allocated to the activities that cause these costs, and then the costs of each activity are distributed among the products, so that each product bears its actual environmental costs. The largest portion of the environmental costs is borne by the ammonia product, as most of the reactive processes, chemical use, and water use occur in the ammonia plant, while the urea plant has some processes for producing urea. This leads to accurate cost calculations, which in turn is reflected in decisions. Management optimizes the use of resources, which in turn reduces costs for the company. The overall profit margin reached 51.8% of the product's selling price. The reason for this increase in product profitability is the use of the ABC system, which charges each product for its actual consumption of environmental resources, whether directly or through the consumption of resources from auxiliary activities.

The ABC system contributed to the redistribution of production costs in a way that enables the company's management to determine the true cost of the product, including environmental costs. It also helped identify the variation in environmental resource consumption for each product individually, given the variation in activity direction. Providing this information will enable management to plan and monitor activities that cause environmental costs within the company. This all contributes to supporting environmental sustainability by conserving resources through the careful use of resources without waste, thus preserving the share of those resources for future generations.

Thus, the research hypothesis has been proven (Applying ABC reduces waste in materials and energy, which contributes to improving decision-making efficiency, enhancing environmental performance, implementing ISO 14001, 140051, and GRI 300 standards, and reducing costs in a way that achieves environmental sustainability).

By compiling physical and financial information, management gains a complete picture of the effects of the company's activities on the environment, as well as the relationship with economic performance, by achieving transparency in the flow of materials within the production process. This supports the proof of the main hypothesis.

Figure 2 shows the steps for applying the EABC technique.

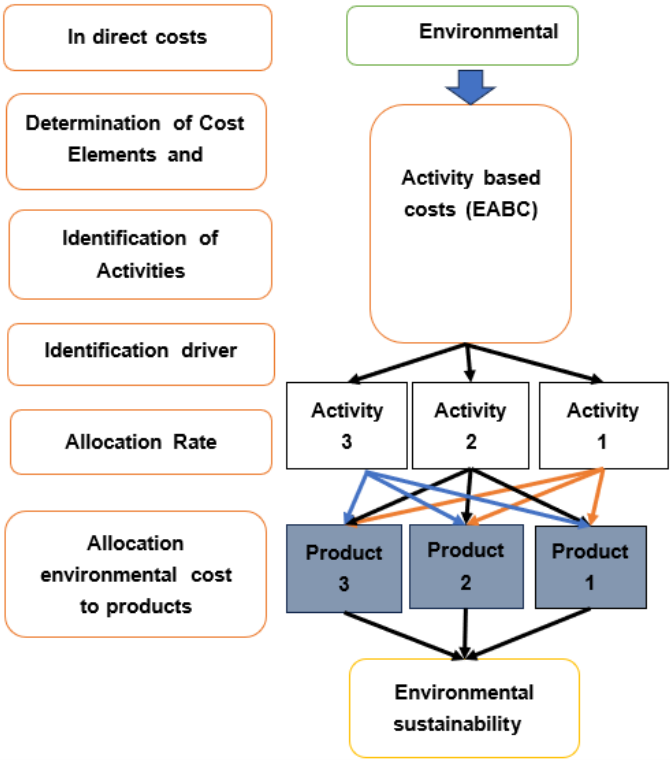


Figure 2. Steps for applying the environmental activity-based costing (EABC) technique

Source: Prepared by the researcher.

5 Discussion

After applying the MFCA flow technique to each unit, the loss of material and waste quantities is accepted according to the ISO 14051 standard and the GRI 300 standard. This is done by comparing the actual inputs with the standard inputs. If the actual inputs are more than the standard level, it causes a waste of those resources. The highest increase in waste in the water unit is in aluminum sulfate, while in the cooling unit it is in chlorine gas, while in the steam boiler unit it is in hydrazine, while in the ammonia unit it is in the reaction gas, while in the urea unit it is in electricity, then water with steam, and the furthest is carbon dioxide gas. Also, after applying the

EABC technique, some important matters are that what is important is what distinguishes this technique from the traditional system, and the traditional method is the disclosure of social costs and hidden within indirect industrial costs. These costs were then allocated across all activities, with each product contributing a small cost. This relies on a reliable strategy to improve environmental performance and reduce costs if this technology is implemented within the company. Brazilian indirect costs, measured using the EABC technology, amounted to IQD 13.88 billion. Each quantity of wasted chemicals was ranked according to its greatest environmental impact per unit, facilitating the maintenance of production efficiency. There are appropriate costs associated with chemical waste due to the nature of the materials themselves. Excessive use of flexible quantities leads to waste beyond what is needed, ultimately increasing chemical waste and negatively impacting the environment. This is because the quantities of these materials are often diverted for other purposes. The EABC technology contributed to reducing the cost of the final product by IQD 68.33 thousand per ton. This is indeed an automatic conversion of non-audio sounds.

Efficiency and effectiveness are among the most important environmental performance indicators used to promote the use of environmentally friendly materials instead of harmful ones. This doesn't preclude the use of raw materials, energy, and fuel, but these must be environmentally friendly and used efficiently. This, in turn, helps reduce environmental costs. Since the company relies on manual additions, resulting in large quantities of materials, it would be better to use a pump to dispense materials in carefully calculated and controlled proportions to avoid resource waste. This, in turn, increases the company's profitability, improves its performance and decision-making, and gives it a competitive advantage for long-term market survival. It also helps conserve resources, ensuring their sustainability for future generations, and reduces environmental damage by decreasing greenhouse gas emissions and the carbon footprint.

6 Conclusion and Recommendations

6.1 Conclusion

The study concluded that the application of the EABC system at the General Company for Fertilizer Industries clearly contributed to identifying and measuring environmental costs more accurately, and linking them to the activities that cause them.

Interviews and field observations have shown that this system provides accurate information that helps management optimize resource utilization and reduce waste and harmful emissions. The importance of the research is highlighted by the fact that it highlights how to support environmental sustainability through modern accounting methods.

The system makes it possible to objectively assess the environmental impact of industrial processes, facilitating decision-making that reduces waste and improves resource efficiency. The results indicate that the adoption of this system contributes to enhancing compliance with international environmental standards such as ISO 14005, 14001, and GRI 300, and reduces the negative impact on the surrounding environment.

It also enables companies to improve environmental performance along with financial performance, supporting their long-term sustainability.

The research shows that the EABC system is not only about improving economic efficiency, but is also a powerful tool for managing environmental impact, reducing waste, and promoting the environmental responsibility of public industrial companies.

6.2 Recommendations

In light of the above, the study recommends the following:

1. Updating accounting systems in Iraqi industrial companies in line with technical and environmental developments, ensuring the provision of accurate information on product costs and their environmental impacts to support management decisions.
2. Adopting an ABC system for allocating environmental costs as the most appropriate method for measuring and charging environmental performance costs, while enhancing cooperation between managers and technicians in different departments to achieve environmental management and economic efficiency.
3. Work to train and qualify administrative and technical personnel in the field of environmental cost accounting to ensure that they perform their duties with high efficiency.
4. Adopt modern environmental management methods that aim to develop environmental protection activities and reduce environmental costs, such as the application of comprehensive environmental management systems in accordance with ISO 14051 and 14001 specifications and GRI300 standards, thereby enhancing the competitiveness of Iraqi companies.
5. Encourage future studies to evaluate and develop environmental management systems and apply modern cost techniques in Iraqi industrial companies, thereby helping to overcome current shortcomings and keep pace with global developments in the field of environmental management.

Author Contributions

Conceptualization, A.A.A.M.A.-T.; methodology, A.A.A.M.A.-T.; software, A.A.A.M.A.-T.; validation, A.A.A.M.A.-T. and E.A.S.A.-M.; formal analysis, A.A.A.M.A.-T.; investigation, A.A.A.M.A.-T.; resources, A.A.A.M.A.-T.; data curation, A.A.A.M.A.-T.; writing—original draft preparation, A.A.A.M.A.-T.; writing—review and editing, E.A.S.A.-M.; visualization, A.A.A.M.A.-T.; supervision, A.A.A.M.A.-T.; project administration, A.A.A.M.A.-T. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflict of interest.

References

- [1] H. S. Abd-Allah and M. R. Abd Alkadm, “Using on green activities based costs to measure costs and improve management decisions,” *Int. J. Res. Soc. Sci. Humanit.*, vol. 13, no. 1, pp. 120–128, 2023. <https://doi.org/10.37648/ijrssh.v13i01.011>
- [2] K. M. Yang and S. H. Chang, “Application of activity-based costing to green industry for profitability and performance enhancement-recycling of blast furnace slag as an example,” *Adv. Manage. Appl. Econ.*, vol. 8, no. 3, pp. 27–59, 2018.
- [3] N. E. John, P. Nwaguru, and A. J. Williams, “Assessing materials vendor selection in construction project supply chain: The relative importance index approach,” *Int. J. Constr. Supply Chain Manage.*, vol. 12, no. 2, pp. 32–46, 2022.
- [4] C. L. Hsieh, W. H. Tsai, and Y. C. Chang, “Green activity-based costing production decision model for recycled paper,” *Energies*, vol. 13, no. 10, p. 2413, 2020. <https://doi.org/10.3390/en13102413>
- [5] I. A. Al Mashkoor, “The impact of green activity-based costing and green supply chain practices on environmental performance of oil refineries in Iraq,” *Int. J. Econ. Finance Stud.*, vol. 14, no. 4, pp. 96–113, 2022.
- [6] S. S. Dhari and M. A. Nouri, “The role of value stream in measuring environmental costs and reducing costs,” *J. Econ. Adm. Sci.*, vol. 30, no. 141, pp. 595–610, 2024. <https://doi.org/10.33095/ftgg5m71>
- [7] A. Awwad, A. L. M. Anouze, and S. Elbanna, “Green product innovation: Influences on environmental sustainability performance,” *Manage. Decis.*, vol. 64, no. 3, pp. 935–958, 2026. <https://doi.org/10.1108/MD-06-2024-1366>
- [8] L. Ivascu, A. Domil, M. Sarfraz, O. Bogdan, V. Burca, and C. Pavel, “New insights into corporate sustainability, environmental management and corporate financial performance in the European Union: An application of VAR and Granger causality approach,” *Environ. Sci. Pollut. Res.*, vol. 29, no. 55, pp. 82 827–82 843, 2022. <https://doi.org/10.1007/s11356-022-21642-8>
- [9] D. A. G. Gaviria, E. M. C. Martínez, O. C. C. Polo, and J. O. S. Arcila, “Accounting mechanism to measure the environmental impact of industries in the Aburrá Valley and its application of the general disclosure requirements related to financial information on sustainability (IFRS S1) and climate-related disclosures (IFRS S2),” *Rev. Gestao Soc. Ambient.*, vol. 17, no. 5, pp. 1–16, 2023. <https://doi.org/10.24857/rgsa.v17n5-013>
- [10] Q. L. Huynh and V. K. Nguyen, “The role of environmental management accounting in sustainability,” *Sustainability*, vol. 16, no. 17, p. 7440, 2024. <https://doi.org/10.3390/su16177440>
- [11] M. A. Mohsin and H. H. Mohammed, “Importance of implementing material flow cost accounting technique (MFCA) and its role to enhance environmental sustainability in industrial companies: An analytical study on a set of refining companies/Erbil,” *J. Account. Financ. Stud.*, vol. 20, no. 71, pp. 491–504, 2025.
- [12] H. Al-Mawali, “Environmental cost accounting and financial performance: The mediating role of environmental performance,” *Accounting*, vol. 7, no. 3, pp. 535–544, 2021. <https://doi.org/10.5267/j.ac.2021.1.005>
- [13] N. H. A. Al-Shaibani, “Using material flow cost accounting technique to reduce environmental costs: An applied study,” *Acad. Account. Financ. Stud. J.*, vol. 28, no. 2, pp. 1–17, 2022.
- [14] V. Vijayalakshmi and M. Harishkumar, “A study on the significance of emerald accounting,” *M.Com Online J.*, vol. 2, no. 1, pp. 92–100, 2019.
- [15] O. Zaim, “Green supply chain practices and their role in enhancing cost efficiency and competitiveness,” *J. Policy Options*, vol. 6, no. 1, pp. 23–27, 2023.

- [16] T. K. Torban, M. E. Jalata, and C. D. Geleta, "The role of institutional pressure, green procurement and reverse logistics practices on environmental performances: Evidence from Ethiopia industrial parks," *Int. J. Ind. Eng.*, vol. 34, no. 3, pp. 1–21, 2023. <https://doi.org/10.22068/ijiepr.34.3.10>
- [17] C. Shao, S. Chen, and X. Zhan, "Sustainable development agenda: Historical evolution, goal progression, and future prospects," *Sustainability*, vol. 18, no. 2, p. 948, 2026. <https://doi.org/10.3390/su18020948>
- [18] S. A. Rehman Khan and Z. Yu, "Assessing the eco-environmental performance: A PLS-SEM approach with practice-based view," *Int. J. Logist. Res. Appl.*, vol. 24, no. 3, pp. 303–321, 2021. <https://doi.org/10.1080/13675567.2020.1754773>
- [19] A. Al-Hebe and E. H. Ali, "The role of the green economy in achieving environmental sustainability," *J. Leg. Econ. Res.*, vol. 15, no. 92, pp. 1–3, 2025. <https://doi.org/10.21608/mjle.2025.454441>
- [20] F. Özçelik, "Integration of activity-based costing to the environmental management accounting," *N. Econ. Trends Bus. Oppor.*, p. 63, 2020.
- [21] W. H. Tsai, S. C. Chang, and Y. Teng, "Equilibrium between environmental and economic objectives: An activity-based costing approach application for carbon emissions management in the aluminum alloy wheel industry," *Energies*, vol. 17, no. 6, p. 1331, 2024. <https://doi.org/10.3390/en17061331>
- [22] M. M. Rahman, M. S. Rahman, and B. C. Deb, "Competitive cost advantage: An application of environmental accounting and management approach with reference to Bangladesh," *Cost Manage.*, vol. 49, no. 3, pp. 47–59, 2021.